

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modelling

Week 13: Stochastic Processes & Brownian Motion

Day 86: Introduction to Stochastic Processes

1 Introduction

In financial markets, prices evolve over time in an uncertain and random manner. To model such behavior, we use **stochastic processes**.

A stochastic process is a collection of random variables indexed by time:

$$\{X_t : t \in T\}$$

where each X_t represents the state of the system at time t .

2 Intuition

A stochastic process can be thought of as:

- A random path evolving over time
- A sequence of outcomes driven by randomness

In finance, examples include:

- Stock prices
- Interest rates
- Volatility

3 Discrete vs Continuous Time

Stochastic processes can be classified based on time:

3.1 Discrete-Time Processes

$$X_0, X_1, X_2, \dots, X_n$$

Example: daily stock returns.

3.2 Continuous-Time Processes

$$\{X_t : t \geq 0\}$$

Example: stock prices modeled continuously over time.

4 Types of Stochastic Processes

4.1 Random Walk

A simple stochastic process:

$$X_{t+1} = X_t + \epsilon_t$$

where ϵ_t are independent random variables.

4.2 Markov Process

A process where the future depends only on the present:

$$P(X_{t+1}|X_t, X_{t-1}, \dots) = P(X_{t+1}|X_t)$$

4.3 Martingale

A process with no predictable drift:

$$E[X_{t+1}|\mathcal{F}_t] = X_t$$

This is a key concept in financial modeling.

5 Filtration and Information

We define $\mathcal{F}_t$ as the information available up to time t .

A stochastic process evolves with respect to this information set.

6 Expectation and Variance

Each random variable X_t has:

$$E[X_t], \quad Var(X_t)$$

These describe the distribution of the process at time t .

7 Why Stochastic Processes in Finance?

Financial markets are inherently uncertain.

Deterministic models cannot capture:

- randomness in price movements
- volatility fluctuations
- unpredictable shocks

Stochastic processes provide a mathematical framework to model these features.

8 From Discrete to Continuous Models

Many financial models start with discrete-time processes and move to continuous-time limits.

This leads to models such as Brownian Motion, which form the foundation of modern quantitative finance.

9 Conclusion

Stochastic processes are the foundation of modeling randomness over time.

They provide the framework for advanced topics such as:

- Brownian Motion
- Ito's Lemma
- Stochastic Differential Equations

These tools are essential for modern quantitative finance.